

MAINT-AI Synthetic Maintenance Manual

Industrial Induction Motor — IM-450

Document Notice

Synthetic demonstration document created for the MAINT-AI Sprint 2 RAG prototype. It is not a manufacturer-issued service manual.

1. Equipment Overview

Equipment: Industrial Three-Phase Induction Motor

Model: IM-450

Typical application: conveyor drives, pumps, and industrial machinery

Normal operating status: Operational

2. Routine Maintenance

Before scheduled inspection, isolate electrical power and follow the site's lockout/tagout procedure. Inspect cooling vents for blockage, check terminal connections for looseness, inspect bearings for abnormal noise, and verify lubrication according to the maintenance schedule.

3. Fault: MOTOR-OVHT-01 — Motor Overheating

Symptoms: motor casing temperature is above the normal operating range, thermal protection may trip, and the motor may produce an abnormal smell.

Possible causes include excessive mechanical load, blocked ventilation, bearing friction, incorrect supply conditions, and insufficient lubrication.

4. Troubleshooting: MOTOR-OVHT-01

1. Stop the motor if the temperature reaches the site's critical shutdown threshold.
2. Verify the mechanical load and check for a jammed driven component.
3. Inspect cooling vents and remove accumulated dust or debris after isolation.
4. Check bearing condition for abnormal noise or friction.
5. Verify lubrication against the scheduled maintenance requirement.
6. Check electrical supply conditions and motor connections.

5. Fault: MOTOR-VIB-02 — Excessive Vibration

Possible causes include misalignment, loose mounting bolts, bearing wear, rotor imbalance, or coupling problems. Inspect mounting hardware and coupling alignment after the equipment is safely isolated. If bearing damage or rotor imbalance is suspected, escalate to qualified maintenance personnel.

6. Safety

Do not open electrical enclosures or inspect rotating components while energized. Use appropriate isolation and lockout/tagout procedures. Do not restart equipment until guards are installed and personnel are clear of the hazard area.